

MATH 2101 LINEAR ALGEBRA I, FALL 2025 –ASSIGNMENT 6
OUTLINE OF SOLUTION

Due date: 14th November (Monday), 2025 10:00pm.

- To receive *full credits*, the solution has to be *clear* and provide sufficient explanations.
- All solutions have to be turned in HKU Moodle in the format of *PDF file*.
- Please include your *Name, UID, Faculty, Major (if declared)* in your solution.

1. Let $\mathcal{P}_5$ be the vector space of polynomials of degree less than or equal to 5. Define $T_1 : \mathcal{P}_5 \rightarrow \mathcal{P}_5$ given by: for any $f \in \mathcal{P}_5$,

$$T_1(f)(x) = f(2x + 1)$$

and

$$T_2(f)(x) = 2f(x) + 1.$$

Determine whether T_1 and T_2 are linear transformations.

Sol. We claim that T_1 is a linear transformation and T_2 is not. For any $f, g \in \mathcal{P}_5$, we notice that

$$T_1(f + g)(x) = (f + g)(2x + 1) = f(2x + 1) + g(2x + 1) = T_1(f)(x) + T_1(g)(x),$$

i.e.,

$$T_1(f + g) = T_1(f) + T_1(g).$$

In the meantime, for any scalar $c \in \mathbb{R}$, one has

$$T_1(cf)(x) = (cf)(2x + 1) = c \cdot f(2x + 1) = c \cdot T_1(f)(x) = (c \cdot T_1(f))(x),$$

i.e.,

$$T_1(cf) = c \cdot T_1(f).$$

Hence, T_1 is a linear transformation.

Now, for T_2 , we notice that for any $f \in \mathcal{P}_5$

$$T_2(2f)(x) = 4f(x) + 1,$$

$$2T_2(f)(x) = 2(2f(x) + 1) = 4f(x) + 2.$$

So $T_2(2f) \neq 2T_2(f)$, T_2 is not a linear transformation.

2. Let $T : V \rightarrow W$ be an invertible linear transformation. Prove that there exists a basis β for V and a basis γ for W such that $[T]_{\beta}^{\gamma}$ is the identity matrix.

Proof. Let $v_1, \dots, v_n$ be a basis for V . We claim that by the injectivity of T , $T(v_1), \dots, T(v_n)$ are also linearly independent and so form a basis for W . Once the claim is proved, we let $\beta = \{v_1, \dots, v_n\}$ and let $\gamma = \{T(v_1), \dots, T(v_n)\}$. Then $[T]_{\beta}^{\gamma} = I_n$.

We proceed to prove the claim. We prove a stronger result.

Proposition. Let $T : V \rightarrow W$ be a linear transformation. Suppose T is injective. Then $v_1, \dots, v_r$ in V are linearly independent if and only if $T(v_1), \dots, T(v_r)$ are linearly independent.

Proof of the Proposition We first prove the only if direction. Suppose $v_1, \dots, v_r$ in V are linearly independent. Then we consider

$$a_1 T(v_1) + \dots + a_r T(v_r) = 0$$

for some $a_1, \dots, a_r$. Hence, $T(a_1 v_1 + \dots + a_r v_r) = 0$. By the injectivity of T , $a_1 v_1 + \dots + a_r v_r = 0$. Now the linear independence for $v_1, \dots, v_r$ gives that $a_1 = \dots = a_r = 0$. Hence $T(v_1), \dots, T(v_r)$ are linearly independent.

We now prove the if direction. Suppose $T(v_1), \dots, T(v_r)$ in W are linearly independent. Then we consider

$$b_1 v_1 + \dots + b_r v_r = 0.$$

Then $T(b_1 v_1 + \dots + b_r v_r) = 0$. By the definition of a linear transformation,

$$b_1 T(v_1) + \dots + b_r T(v_r) = 0.$$

The linear independence then gives $b_1 = \dots = b_r = 0$. Hence, $v_1, \dots, v_r$ are linearly independent. $\diamond$

□

3. Let $T' : V \rightarrow W$ and let $T'' : W \rightarrow X$ be linear transformations. Suppose T'' is injective. Prove that $N(T') = N(T'' \circ T')$.

Sol: For $v \in N(T')$, $(T'' \circ T')(v) = T''(T'(v)) = T''(0) = 0$. So $v \in N(T'' \circ T')$. Conversely, for $v \in N(T'' \circ T')$, i.e., $(T'' \circ T')(v) = 0$, the injectivity of T'' implies $T'(v) = 0$. Hence $v \in N(T')$. Hence we conclude that $N(T') = N(T'' \circ T')$.

4. Let M_3 be the vector space of 3×3 matrices. Let

$$P = \begin{pmatrix} 1 & -2 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Define $T : M_3 \rightarrow M_3$ given by:

$$T(A) = PAQ.$$

Find a basis for $R(T)$ and a basis for $N(T)$. Explain your answer. (Hints: Let $T'(A) = AQ$ and $T''(A) = PA$. Consider $T = T'' \circ T'$. What can you tell about

$T''?$)

Sol: We have

$$T'(A) = AQ = \begin{pmatrix} a_{11} + a_{12} + a_{13} & a_{11} + a_{12} + a_{13} & a_{11} + a_{12} + a_{13} \\ a_{21} + a_{22} + a_{23} & a_{21} + a_{22} + a_{23} & a_{21} + a_{22} + a_{23} \\ a_{31} + a_{32} + a_{33} & a_{31} + a_{32} + a_{33} & a_{31} + a_{32} + a_{33} \end{pmatrix}.$$

Then, setting the free variables, $a_{12} = s$, $a_{13} = t$, $a_{22} = u$, $a_{23} = v$, $a_{32} = p$ and $a_{33} = q$. Then $a_{11} = -s - t$, $a_{21} = -u - v$ and $a_{31} = -p - q$. Then a basis for $N(T')$ is:

$$\left\{ \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{pmatrix} \right\}.$$

By Question 3, we have that $N(T'' \circ T')$ has the same basis.

By the dimension formula, $\text{rank}(T'' \circ T') = \dim(M_{3 \times 3}) - \text{nullity}(T'' \circ T') = 9 - 6 = 3$. Similarly, $\text{rank}(T') = 3$. We may guess here. Consider

$$\begin{aligned} T' \left(\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right) &= \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ T' \left(\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right) &= \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \\ T' \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \right) &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}. \end{aligned}$$

It is quite straightforward that the above three vectors are linearly independent and so form a basis for $R(T')$ (we have explained that $\text{rank}(T') = 3$). Now, note that since

$$\det \begin{pmatrix} 1 & -2 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 1 \end{pmatrix} = 1 \neq 0$$

and so P is invertible. Thus T'' is also invertible with the inverse $T''^{-1}(A) = P^{-1}A$.

Then, by Question 2

$$\left\{ T'' \left(\begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right), T'' \left(\begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \right), T'' \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \right) \right\}$$

are linearly independent and so form a basis for $R(T'' \circ T')$ again. A direct computation gives that:

$$\left\{ \begin{pmatrix} 1 & 1 & 1 \\ -1 & -1 & -1 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} -2 & -2 & -2 \\ 3 & 3 & 3 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \right\}$$

forms a basis for $R(T'' \circ T')$.